Methodology & Implementation

1. Theoretical framework

We've structured our index around five theoretical drivers of democratic change: economic inequality and relative deprivation (D1), information environment shocks (D2), elite defection and intra-elite conflict (D3), state capacity erosion (D4), and polarisation and effective tribalism (D5). These drivers were selected on two grounds: each is well-established in the political sciences and comparative democratisation literature as a structural condition associated with democratic instability, and each has a theoretically coherent and constructable pathway through which AI can plausibly amplify the condition. We do not claim that these drivers are exhaustive – for example, religious, demographic, and geopolitical shocks have independent literature spheres — but they represent those drivers able to canvas the widest range of contributors to modern democratic stability and most exposed to AI-specific amplification mechanisms; the latter is the framework's stated analytical focus.

Consider the following descriptions and theoretical bases for each of the five drivers:

Driver	Label	Primary theoretical basis in the literature	Relevance to democratic stability
D1	Economic inequality & relative deprivation	Meltzer & Richard (1981); Piketty (2014); Acemoglu & Robinson (2012)	Material grievance generates redistribution pressure; when that pressure fails to translate into policy, for example, through elite capture, collective action problems, or identity politics, it accumulates as a structural source of political volatility
D2	Information environment shocks	Prat & Strömberg (2013); Guriev & Treisman (2022)	Democratic accountability depends on the quality of the signal voters receive about leaders and policies; structural shocks to the information environment undermine the epistemic preconditions for that accountability, independently of citizens' material conditions
D3	Elite defection & intra-elite conflict	Goldstone (1991); Turchin (2016, 2023)	Democratic arrangements lose their equilibrium not primarily because of mass mobilisation but because the coalition sustaining them fractures from within; the conditions for defection are structural, not ideational

D4	State capacity erosion	Besley & Persson (2011); Fukuyama (2011, 2014)	A state that cannot raise revenue effectively, enforce contracts impartially, or deliver public goods reliably cannot respond to the grievances the other drivers generate; erosion is self-reinforcing and can be deliberate as well as structural
D5	Polarisation & effective tribalism	Iyengar et al. (2019); Abramowitz (2010, 2018)	Sufficiently deep partisan divisions make democratic correction harder independently of policy content; when partisan identity becomes tribal, voters tolerate anti-democratic behaviour by co-partisans that they would condemn in opponents, and cross-partisan coalitions for institutional investment become structurally unavailable

D1 — Economic inequality and relative deprivation is perhaps the most broadly accepted driver of political change in the comparative literature. Meltzer and Richard's median-voter model establishes a self-correcting mechanism whereby, as the distribution of income widens and the median voter's income falls relative to the mean, support for redistribution widens and democratic pressure encourages egalitarian policy correction. Acemoglu and Robinson identify why this correction fails in practice — collective action problems and elite capture prevent the majority preference from translating into tractable policy. Piketty's $r > g$ condition enriches the structural dimension: when returns to capital exceed economic growth, inequality deepens regardless of formal democratic procedures. We treat D1 as operating through relative deprivation rather than absolute poverty, given that the literature consistently links the perceived gap between deprivation and poverty as the link to political volatility rather than material deprivation in and of itself.

D2 — Information environment shows first follows from Prat and Strömberg's premises established in their 2013 work: voters are rationally ignorant and rely on media as a substitute for costly information-gathering, making the quality of the information environment a structural precondition for accountability. Where the media is more plural, more competitive, voter knowledge is better, and policy is more responsive. Guriev and Treisman extend this to its most consequential form, the spin dictator. This individual maintains power not through outright repression but by systematically controlling observable signals to manufacture apparent legitimacy. D2 is structurally distinct from D1 in that economic deprivation creates conditions for democratic correction; information deprivation can mask those conditions entirely or manufacture false ones.

D3 — Elite defection and intra-elite conflict operate at a more concentrated level than D1 and D2. Goldstone's structural-demographic theory identifies three co-moving conditions for state breakdown: state fiscal distress (SFD), elite overproduction (or elite mobilisation potential, EMP), and popular mobilisation (MMP). The elite overproduction mechanism is load-bearing— when aspirants to elite positions (the elite class) grow faster than available positions, disaffected aspirants become available as political entrepreneurs willing to mobilise mass grievances. The mechanism is structural rather than ideational; defection follows from a shift in the payoff structure of loyalty and not from a shift in values. Turchin's political stress indicator formalises this as a quantitative tracker of the same three components. Applying these indicators to contemporary, advanced democracies reveals expanded credentialed classes whose expectations outrun elite absorption as the defining stress condition.

D4 — State capacity erosion is decomposed by Besley and Persson into fiscal and legal components, which are then modelled as stocks requiring active political investment to maintain. The core insight is that this investment is undersupplied under political conflict. Each faction blocks capacity-building that a rival might turn against them, in turn producing a coordination failure when strong institutions are needed most. Fukuyama adds a decay mechanism, repatrimonialisation, or the effect whereby impersonal institutions are progressively captured by insider networks. Further, the vetocracy of accumulated veto players in mature democracies contributes to subtler but equally damaging erosion. It should be noted that D4 has an important asymmetric dimension with respect to AI: should AI increase coercive state capacity while simultaneously degrading accountability capacity, a scalar score will miss this nuanced directional shift.

D5 — Polarisation and effective tribalism is distinguished from ideological polarisation by Iyengar and colleagues as partisan dislike/distrust independent of policy distance (affective polarisation). Critically, affective polarisation has increased sharply in established democracies even while actual policy preferences have not comparably diverged. The divide is social and psychological rather than substantive, and therefore not self-corrective through policy convergence. Its consequence for democratic function is that tribal partisan identity generates tolerance for anti-democratic behaviour by co-partisans, severing the procedural norms on which accountability depends. Abramowitz adds the structural dimension: progressive sorting of racial, cultural, and ideological identities into party alignment has shrunk the ambivalent centre, shifting electoral incentives from persuasion to base mobilisation and reinforcing polarisation from the elite level downward. This driver also acts as a cross-cutting amplifier in that it blocks coalitions that would address the condition the other four drivers generate. Its internal consistency is notably lower than D1–D4 (α = 0.088 for cross-sectional (CS); 0.309 for panel), reflecting genuine construct multidimensionality; this is addressed in section 4.

Figure 0. Causal directed graph

Justifying the inclusion of non-AI-specific indicators

The index includes both AI-specific indicators and non-AI-specific democratic health indicators. The latter are a structural requirement of the analytical framework.

The amplification framing fundamentally requires a baseline to amplify against. Our core analytical claim is that AI amplifies pre-existing structural drivers of instability. Testing an amplification claim requires measuring the driver and the amplifier independently, something that becomes impossible without non-AI baseline indicators (in this case, the composite would measure AI exposure rather than AI-conditioned democratic risk). This is a structural requirement of the regression analyses that test whether higher AI exposure scores predict deterioration in democratic health. Therefore, democratic health must be measured on its own terms, independently of the AI exposure variable.

The drivers have structural histories that predate AI. Anchoring the index in non-AI indicators allows us to inherit the construct validity of established democratic health measures — V-Dem, Freedom House, WJP, EIU, *et cetera* — that have been methodologically refined over many years and that were designed specifically to operationalise the constructs our theoretical framework specifies. Building the index exclusively on AI-specific indicators would require constructing that validity from scratch, with shorter time series, more limited country coverage, and admittedly weaker theoretical anchoring.

These two indicator types serve distinct measurement functions. Non-AI indicators measure the current severity of each driver as a structural condition; AI-specific indicators measure the degree to which AI is present as an amplifier of that condition. Conflating them would make it impossible to distinguish a country with deep structural vulnerability and low AI exposure from one with moderate structural vulnerability and high AI exposure.

This dual structure does not resolve the underlying identification challenge, namely that AI exposure barely varies before 2023 and most cross-sectional variation is between countries rather than within them. This is addressed in the discussion of analytical methods. What it does ensure is that the composite measures what it claims to: democratic risk conditioned by AI exposure, not AI exposure *per se*.

2. Data selection

The criteria for the indicators were relevance and validity. The relevance was assessed by establishing a clear link to our theoretical framework. The validity was assessed by evaluating its source reputation and methodological rigor.

At the initial selection stage, we did not panelise indicators with limited coverage. Initially, we selected all indicators that serve as a close proxy to the constructs in our theoretical frameworks. Two researchers have worked independently to identify suitable indicators, with one covering AI-specific and the other covering demographic health indicators. The table below shows the datasets surveyed in this stage.

AI-specific Indicators	Democratic Health datasets
CAIDP AI & Democratic Values Index	V-Dem
GIRAI - Global Index on Responsible AI	EIU Democracy Index
IMF AI Preparedness Index	Freedom House Freedom in the World

Ipsos AI Monitor, OECD AI Incidents Monitor	Digital Society Project (DSP)
Stanford AI Index	Transparency International CPI
Tortoise Global AI Index	WJP Rule of Law Index
World Bank GovTech Maturity Index (GTMI)	BTI - Bertelsmann Transformation Index
Digital Society Project (DSP)[1]	IBP Open Budget Survey
OECD Truth Quest Survey, Reuters Institute Digital News Report	CIVICUS Monitor
	IDEA GSoD - Global State of Democracy
	RSF Press Freedom Index

Table 1. Data sources

By the end of this stage, we had two types of indicators: 29 AI-specific indicators and 88 non-AI-specific (democracy health indicators). Data integration process ensured consolidating duplicated indicators (e,g four V-Dem Digital Society Project variables present in both the AI and non-AI datasets).

The main challenges with AI indices are the lack of construct validity. The lack of AI-specific indicators made us rely on nearest proxies, which may not capture the underlying concept precisely. This suggests the need for labeling new datasets on observed indicators or expert sources indicators on AI-specific democracy indicators.

Another challenge of using secondary sources is that each of these indices has its own assumptions about the constructions, weighting, and aggregation. Therefore, accounting for the subjective judgment of other researchers and behind these indices, along with our own assumptions, is important to consider while interpreting our results.

In our cross-sectional data from 2023 to 2025, we use the latest value for a country. Note that this varies by country for the same indicator, and some of the indicators, like WJP and WB GovTech, were taken from 2021. These indicators did not have data after 2021, which also raises a methodological limitation, as the observations for a country may correspond to different points in time.

3. Imputation of missing data

For the selected indicators, we performed a missingness audit; the indicator was labelled as either observed in cases where an indicator is reported for the country at least once.

Since there is no test to determine the reasons why indicators are not missing completely at Random, we used logical judgment to classify variables as:

3.1. Missing Not at random (MNAR)

These are variables whose missingness depends on the values themselves. This data is missing because surveys did not include the countries in their sample due to data collection feasibility, project scope, or deliberate decisions by the researchers. In the context of AI indicators, we are assuming that the misingness can be related to limited AI exposure, for example, where countries with limited AI policies and AI infrastructure are not reported in the AI-related indicators because they don't have an attractive AI ecosystem for researchers to study them. For MNAR variables, we did not perform imputation on them to avoid introducing bias (JRC, 2025). In the table below, we list the variables we classify as MNAR and the rationale behind our decision.

Source	Missing countries	Indicators affected
GIRAI - Global Index on Responsible AI	Bangladesh, Denmark, Israel, Sweden	National AI Policy dimension score; Transparency and Explainability dimension score
Ipsos AI Monitor	UAE, Bangladesh, Denmark, Ghana, Israel, Nigeria, Taiwan	% AI will replace jobs.
Reuters Institute (AI sub-report)	All except Denmark, France, Japan, the UK, USA	6 Reuters AI-and-news metrics
OECD Truth Quest (TQ12)	Bangladesh, Chile, China, Denmark, Ghana, India, Indonesia, Israel, New Zealand, Nigeria, Singapore, South Africa, South Korea, Sweden, Taiwan, UAE	TQ12 - Truth Quest score by type: Disinformation (AI-generated)
Stanford	All except Germany, Denmark, France, the UK, the Netherlands, Poland, Sweden, USA	Total Public Spending indicator, Number of AI bills, % Trust AI not to discriminate
OECD AI Incidents Monitor	Bangladesh, Chile, Denmark, Germany, Ghana, Indonesia, Nigeria, Poland, South Africa	AI Incidents count — presence-only system: absence means zero reported, not confirmed zero incidence
BTI (Bertelsmann)	Australia, Canada, Denmark, France, Germany, Israel, Japan, Netherlands, New Zealand, Sweden, UK, USA	BTI by design covers only transformation/developing countries
IBP Open Budget Survey	All except Bangladesh, Brazil, Chile, China, India, Indonesia	IBP samples only select countries per wave
WJP Rule of Law	Israel, Taiwan	These 2 countries are not in the WJP scoring editions

V-Dem election indicators	China	China holds no contested multiparty elections

Table 2. MNAR indicators

3.2. Missing at Random (MAR)

The only indicator that we found to be missing at random, meaning that the missingness does not depend on the variable of interest, was World Bank GovTech. This is because it reports data every two years. The observed waves were in 2020, 2022, and 2025. We used linear imputation to handle these missing values. And as with all imputation techniques, their limitation include underestimating the variance. This is because imputed values were later treated as actual values. We accounted for this in the sensitivity analysis, where we used the actual values with no imputation and found that it has no impact on the final ranking.

After data imputation, we applied a ≤ 30% %threshold for acceptable missingness (≥ 19 / 27 countries). Out of our 118 indicators, 93 indicators pass the threshold, and 25 indicators were deleted. The rationale behind this relatively high threshold is the small sample size (27 countries). This threshold also encourages inclusion, as it avoids including only data-dense countries, as the countries that are included in AI indices may already have qualities that comparative advantage in the AI race, e,g(income level, AI infrastructure, but the downside of this is that useful indicators for the conceptual framework(e,g AI deepfake indicators in the cross sectional setting) won't qualify to contribute to the index of a country.

3.3. Country-level missingness audit

10 out of the 27 countries have missing data in the final selected indicators. This is an important caveat to account for when looking at the scores of these countries. As you will see in our weighting, we used partial weighting for countries with missing values. The country missingness audit is reported in the table below.

Country	Missing Count	Percentage Missing	CORDA Rank
Israel	10	12.20%	16
Bangladesh	8	9.76%	26
Denmark	7	8.53%	1
Taiwan	6	7.32%	11
Sweden	5	6.10%	2
Ghana	3	3.66%	19
United Arab Emirates	3	3.66%	25
Nigeria	3	3.66%	24
China	3	3.66%	27

Indonesia	1	1.22%	21

Table 3. Country-level missingness

4. Multivariate analysis

Cronbach's alpha was used to measure the eligibility of indicators in a dimension by measuring how well the indicators measure the same unidimensional object(JRC, 2005). A 0.7 threshold was used to determine the eligibility of an indicator. Cronbach's alpha can be calculated as follows

$$\alpha = Q/(Q-1) \times (1 - \Sigma\, Var(x_j) / Var(x_0))$$

Where:

- Q = The total number of indicators/items in the driver
- x_j = The score of indicator j
- x_0 = The total combined score across all indicators from 1 up to j (Σx_j)
- $Var(x_j)$ = The variance of indicator j
- $Var(x_0)$ = The variance of the total combined score across all Q indicators

We performed the test on both cross-sectional and panel data. The table below presents the result.

Driver	CS N complete	Panel N	α (CS)	Interpretation
D1 Economic Inequality	17	351	0.941	High internal consistency with possible redundancy
D2 Information Environment	20	81	0.943	High internal consistency with possible redundancy
D3 Elite Defection	21	0	0.920	High internal consistency with possible redundancy.
D4 State Capacity	19	0	0.957	High internal consistency with possible redundancy.
D5 Polarisation	27	459	0.088	Low internal consistency, suggesting its separation in the panel phase

Table 4. Internal consistency results

To understand how indicators within a dimension relate to one another, we calculated the correlation of indicators with one another. The formula used is shown below:

$$r\_XY = Cov(X,Y) / (\sigma\_X \times \sigma\_Y)$$

Where

Cov(X, Y) = the covariance between variable X and variable Y. It shows whether X and Y tend to move together.

σ_X = standard deviation of X

σ_Y = standard deviation of Y

For indicator pairs with an r value of more than 0.9, we assume the potential of redundancy and assess the potential of consolidation. Below, we present the number of redundant pairs per driver

Driver	Redundant pairs
D1 Economic Inequality	2
D2 Information Environment	15
D3 Elite Defection	18
D4 State Capacity	29

Table 5. Redundant pairs (r>0.9) within drivers

Based on the analysis, in Driver 1, we removed *v2xeg_eqprotec*: Equal Protection Index and retained *v2x_egal*: Egalitarian Component Index because it represents the composite broader score, part of which is *v2xeg_eqprotec*: Equal Protection Index.

In Driver 4, we removed CIVICUS: Civic Space Rating and retained CIVICUS: Civic Space Score because they measure the same thing in different scales.To avoid multicollinearity, we removed WJP: Factor 2, 6, 7, 8 sub-factors and retained the composite indicator WJP: Rule of Law Index Overall Score

We are careful not to overinterpret these results, as a high alpha coefficient value may not necessarily measure unidimensionality; instead, it may happen because of highly correlated indicators(JRC, 2025). The limitation of our correlation analysis is that we compare indicators within a driver, but we do not compare indicators across drivers. While running statistical methods of clustering the indicators can theoretically help us verify our clustering, the limited sample size makes it unreliable.

5. Normalisation

The need for normalisation comes from the fact that indicators have different measurement units, and they may also be normalised to different scales. As a result, indicators with a larger scale will have more weight than an indicator with a smaller scale in the aggregated index score. This is relevant for our dataset, where each of the indicators we use has different scales, e.g., some have a scale of 1-100, and others have a scale of 0-1. Min-max normalisation also maintains the relationship between values proportionally by shrinking the distribution within the new assigned minimum and maximum values. The normalised values can be calculated as follows:

$$X' = (X - min(A)) / (max(A) - min(A)) \times (max' - min') + min'$$

Where:

X' = the normalised or rescaled value

X = the original value

$min(A)$ = the smallest value in the original dataset or indicator

$max(A)$ = the largest value in the original dataset or indicator

max' = the highest value in the new scale, which is 0.

min' = the lowest value in the new scale, which is 100.

Not that the global minimum and maximum were computed across all countries and all years to allow temporal comparability of scores. For negative values, the observed global minimum was used regardless of sign. This is because negative values (e,g in all V-Dem/DSP Bayesian interval estimates) are not suitable for min-max normalisation

Before normalising the values, we unifyed the direction of all indicators. Therefore, we reversed the scale for indicators that are associated with democratic risks so that a higher number will always indicate better democratic health.

Indicator	**Why reversed**
% AI Will Replace Jobs (Ipsos)	A higher score means more agreement, indicating public concern about AI job displacement and, therefore, high exposure
OECD AI Incidents Monitor	A higher count of AI deepfake incidents relates to high exposure to information shocks
AI Companies (Tortoise)	A composite indicator that measures how developed a country’s AI sector is. A higher value may indicate high AI risk exposure.
AI Funding (Tortoise)	A composite indicator that assesses the funding in AI, where a high value indicates higher funding, which may indicate the possibility of creating Economic elites relative to existing elites.
RSF Press Freedom Score	A lower number refers to more free press
v2x_corr: Political Corruption Index	A higher score means more corruption
v2x_neopat: Neopatrimonial Rule Index	A higher score means more neopatrimonial rule
v2caautmob: Mobilisation for	A higher score means more mobilisation for

Autocracy	autocracy, which indicates higher autocracy initially
v2cacamps: Political Polarisation	A higher score means higher polarisation

Table 6. Reversed indicators

6. Weighting and aggregation

We applied equal weighting across drivers, with each driver contributing with 20%. Within each driver, we applied partial weighting, and the weight of each indicator is equal to 1/n, where n is the number of available indicators. The table below lists the weight assigned to each driver and

The indicators within the driver.

Driver	Driver weight	Indicators per driver	Weight per indicator
D1 Economic Inequality	20%	9	11.11%
D2 Information Environment	20%	21	4.76%
D3 Elite Defection	20%	21	4.76%
D4 State Capacity	20%	22	4.55%
D5 Polarisation	20%	9	11.11%

Table 7. Weighting details

An important note regarding partial weighting is that in the case of missing data per indicator, the weight becomes 0. We apply this to avoid panelising countries for missing data. The limitation for this decision is that, depending on the missing data, it either penalises or flattens the data. For example, if an indicator value is associated with democratic risk exposure and it does not have a weight because the country does not have a reported value for it, the final score will overestimate the country's readiness. In cases where the indicator value associated with readiness and the country (if reported) may have a high value, then partial weighting will panelise the country. In either case, this decision creates compatibility constraints between countries with different counts of missing data, as the table of the country-level missingness shows. Moreover, due to the variation in the number of indicators per driver, the weight for each individual indicator within a driver depends on the number of indicators within that driver. In other words, indicators in driver 1 and driver 5 will have a stronger influence over the index score than indicators in driver 2,3 or 4.

7. Uncertainty and sensitivity analysis

In accounting for the uncertainty around the design decisions related to imputation, normalisation, and weighting, we tested different variants of the index and compared them with our original index. For MAR imputation, instead of using interpolation, we deleted the columns and observed the difference in ranking. Also, we constructed a variant of the index without Ipos indicators, as they had the highest missingness (20/27) in the final selected indicators, and we compared the resulting rankings against our final ranking. We found that the index ranks were fully robust(Kendall τ = 1.0000) to changing imputation. For normalisation, we tried using Z scores instead of min-max normalisation. The ranking was robust and insensitive (Kendall τ = 0.9829) to changing normalisation methods.

For weighting, apart from equal weighting (Specification A), we defined two different specifications, each with different weighting decisions.

The specifications we used to account for uncertainty regarding weighting are:

1. Specification A: Equal weights to all drivers. Each driver contributes by 20%
2. Specification B: Assign a higher weight (30%) to driver 2: information environment shocks relative to the other drivers(each weighting 17.5%), assuming that its impact is higher than the other drivers.
3. Specification C: Assign a higher weight to driver 4: state capacity erosion (30%) relative to other drivers, each contributing by 17.5% to the final score. This assumes that the capacity of the state should be given more attention when assessing the impact of AI on democratic aspects.
4. We also used PCA weights[2].

When we compared the ranking of the different specifications compared the ranking with our equal weighting ranking. In all these different specifications, the index ranks were stable and robust(Kendall τ between 0.9487 and 0.9829).

For aggregation, we note that the major limitation of linear aggregation is that it allows for compensation, meaning a high score in one driver would compensate for a low score in another driver. The geometric mean solves this by panelising for variation in driver scores (JRC, 2005). This is why when we implemented geometric aggregation, the rank of New Zealand changed from 10 to 19 (Kendall τ = 0.7835). New Zealand's score on driver 5 (Polarisation/Affective Tribalism) is 51.8, while its other four drivers range from 67 to 85.

All these sensitivity tests revealed the high robustness of our index, except for the aggregation test, which revealed some instability for New Zealand and somewhat for Sweden. The table below shows the sensitivity test, *Kendall_tau*, and the number of countries whose index shifted by 5.

Test	**Kendall τ vs primary**	**Countries shifting >5 ranks**
T1: Z-score vs min-max normalisation	0.9829	0
T2: Geometric vs linear aggregation	0.7835	1 (New Zealand)

T3b: Spec B — D2=30% (Info-heavy)	0.9487	0
T3c: Spec C — D4=30% (Capacity-heavy)	0.9829	0
T4: No imputation (WB GovTech observed only)	1.0000	0
T5: Borderline Ipsos indicators excluded	1.0000	0
T6 (new): PCA weights vs equal weights	0.9715	0

Table 8. Sensitivity test results

[2] It's important to note that PCA performance is not stable for Driver 3 and Driver 4 due to sample size, and when applied for Panel data on Driver 1 and Driver 2, it excludes AI-only indicators

8. Links to other indicators

In the table below, we compare our index with different AI governance indicators, namely, Oxford Insights GIARI, CAIDP. These indicators were chosen out of the other indices because they show the closest link to assessing countries' readiness to democratic risks.

Index	Oxford AI insights GIRAI	CAIDP
Focus	Government AI readiness	Human rights-based benchmark to assess countries and capacities
Dimensions	Policy capacity, AI governance, AI infrastructure, Public sector adoption, diffusion, and Resilience.	Responsible AI capacities, Human rights, and responsible AI governance
		They have 12 questions, each of which is scored either (Y, N, P). The scores are reported based on an analysis of a country's AI commitments and governance
What's missing?	It focuses on the government's readiness rather than societal readiness. In their theoretical framework, the goodwill of the state is assumed. For us,	It focuses on preparedness rather than exposure, and even under readiness, the AI impact on the information environment and the integration of AI in the public

	this means the readiness of authoritarian governments to utilise AI, creating high AI democratic risks. In its framework, the index also assumes that AI adoption in the public sector can correlate with government readiness for AI. This does not account for AI eroding state capacity due to gradual decision-making automation. The optimistic view is also seen in the AI infrastructure, as we think they may contribute to AI power concentration.	sector have not been addressed. It assumes that the current AI policies account for democratic risks from AI (e.g. economic equality, information environment). The UDHR may give insights into a country's propensity to misuse AI, but for countries with visible UDHR commitment, AI power concentration may lead to new forms of human rights violations. Moreover, the limited scale for assessing countries' commitment and implementation is small (3 values only) which makes it hard to identify small differences among countries.

Table 9. Comparison with other indicators

9. Visualisation of the results

Tool interface overview

The interface enables users to customise their view based on their preferences and interests.

The image below demonstrates the different components in our interface

Figure 1. CORDA Dashboard filter

1. Main components: data views:

    a. Ranking in stacked bar charts

Figure 2. CORDA rankings stacked bar view

    b. Two-axis scatter plot: compare different drivers with each other or with the overall score

Figure 3. CORDA rankings scatter plot view

c. World map

The world map view allows users to press a country's position in the world map and review its respective score. The darkness of the color corresponds to the democratic health level, with darker navy indicating good performance. Users can choose whether they want to view the composite score or the driver level scores.

Figure 4. CORDA world map view

Note on AI usage

Large language models were used to help in extracting indicators for initial screening, which may introduce biases because of the representation of their training data. Also, Claude Cowork was used in data preprocessing ,verification and dashboard coding using a customised pipeline.

References

JRC / Nardo, M., Saisana, M., Saltelli, A., & Tarantola, S. (2005). Tools for Composite Indicators Building. EUR 21682 EN. European Commission Joint Research Centre. https://publications.jrc.ec.europa.eu/repository/handle/JRC31473.

The OECD Composite Index checklist

Step	Why is it needed	
1. Theoretical framework	Provides the basis for the selection and combination of variables into a meaningful composite indicator under a fitness-for-purpose principle (involvement of experts and stakeholders is envisaged at this step)	• To get a clear understanding and definition of the multidimensional phenomenon to be measured. • To structure the various sub-groups of the phenomenon (if needed). • To compile a list of selection criteria for the underlying variables, e.g., input, output, process.
2. Data selection	Should be based on the analytical soundness, measurability, country coverage, and relevance of the indicators to the phenomenon being measured, and the relationship to each other. The use of proxy variables should be considered when data are scarce (involvement of experts and stakeholders is	• To check the quality of the available indicators. • To discuss the strengths and weaknesses of each selected indicator. • To create a summary table on data characteristics, e.g., availability (across country, time), source, type (hard, soft, or input,

	envisaged at this step).	output, process)
3. Imputation of missing data	Is needed in order to provide a complete dataset (e.g. by means of single or multiple imputation)	• To estimate missing values. • To provide a measure of the reliability of each imputed value, so as to assess the impact of the imputation on the composite indicator results. • To discuss the presence of outliers in the dataset.
4. Multivariate analysis	Should be used to study the overall structure of the dataset, assess its suitability, and guide subsequent methodological choices (e.g., weighting, aggregation)	• To check the underlying structure of the data along the two main dimensions, namely individual indicators and countries (by means of suitable multivariate methods, e.g., principal components analysis, cluster analysis). • To identify groups of indicators or groups of countries that are statistically "similar" and provide an interpretation of the results. • To compare the statistically determined structure of the data set to the theoretical framework and discuss possible differences
5. Normalisation	Should be carried out to render the variables comparable	• To select suitable normalisation procedure(s) that respect both the theoretical framework and the data properties. • To discuss the presence of outliers in the dataset as they may become unintended benchmarks. • To make scale adjustments, if necessary. • To transform highly skewed indicators, if necessary.
6. Weighting and aggregation	Should be done along the lines of the underlying theoretical framework.	• To select appropriate weighting and aggregation procedure(s) respect both the that theoretical framework and the data properties. • To discuss whether correlation issues among indicators should be accounted for. • To discuss whether compensability among indicators should be allowed.
7. Uncertainty	Should be undertaken to assess the	• To consider a multi-modelling

and sensitivity analysis	robustness of the composite indicator in terms of e.g., the mechanism for including or excluding an indicator, the normalisation scheme, the imputation of missing data, the choice of weights, the aggregation method.	approach to build the composite indicator, and if available, alternative conceptual scenarios for the selection of the underlying indicators. • To identify all possible sources of uncertainty in the development of the composite indicator and accompany the composite scores and ranks with uncertainty bounds. • To conduct sensitivity analysis of the inference (assumptions) determine what sources and of uncertainty are more influential in the scores and/or ranks.
8. Back to the data	Is needed to reveal the main drivers for an overall good or bad performance. Transparency is primordial to good analysis and policymaking.	• To profile country performance at the indicator level so as to reveal what is driving the composite indicator results. • To check for correlation and causality (if possible). • to identify if the composite indicator results are overly dominated by few indicators and to explain the relative importance of the sub-components of the composite indicator.
9. Links to other indicators	Should be made to correlate the composite indicator (or its dimensions) with existing (simple or composite) indicators as well as to identify linkages through regressions.	• To correlate the composite indicator with other relevant measures, taking into consideration the results of sensitivity analysis. • To develop data-driven narratives based on the results.
10. Visualisation of the results	Should receive proper attention, given that the visualisation can influence (or help to enhance) interpretability	• To identify a coherent set of presentational tools for the targeted audience. • To select the visualisation technique that communicates the most information. • To present the composite indicator results in a clear and accurate manner.

[1] DSP appear twice because there was a structural duplication in the indicators used for D5. It was later resolved in data integration step.